

Rafael Barbedo

Hydrologist | Geospatial Data Scientist | Research Software Engineer

[GitHub](#) | [LinkedIn](#)

rbarbedofontana@gmail.com | Porto Alegre, Brazil

Skills

- **Hydrological modelling:** Large-scale hydrological modelling, flood forecasting, water availability analysis, and reference streamflow estimation.
- **Geospatial data science:** Processing and analysis of topography, remote sensing, land-use, LiDAR, and catchment-scale spatial datasets.
- **Environmental data engineering:** Reproducible pipelines for environmental time series, geospatial databases, quality control, validation, and data products.
- **Scientific computing:** Scientific software development with Python, SQL, JavaScript, Fortran, statistical modelling, machine learning, and deep learning.

Tools: Python, PostgreSQL, SQLite, VS Code, Linux, Docker, JavaScript, SQL, Fortran, NetCDF/Zarr

Experience

Postdoctoral Researcher

Mar 2026 – present | [Federal University of Rio Grande do Sul \(UFRGS\)](#) in partnership with the [National Water and Sanitation Agency \(ANA\)](#)

- Large-scale hydrological modelling and environmental data analysis for national water resources applications in Brazil.
- Assessment of anthropogenic influences and climate-change impacts on water systems.
- Modelling work with the Modelo de Grandes Bacias (MGB) large-basin hydrological model.

Postdoctoral Researcher

Jul 2025 – present | [Federal University of Rio Grande do Sul \(UFRGS\)](#) in partnership with the [Brazilian Geological Survey \(SGB\)](#)

- Processing and analysis of geospatial and time-series data for flood early warning systems in Rio Grande do Sul.
- Water level and streamflow modelling, quality control, validation, and dashboard support.

Hydrology and geospatial data consultant

May 2025 – Nov 2025 | [Brazilian National Water Agency \(ANA\)](#)

- Modelled and structured a relational and geospatial PostgreSQL/PostGIS database for water availability datasets used by the National Water and Sanitation Agency (ANA).
- Developed data modelling, migration routines, processing functions, documentation, and technical training material.

Hydrological modelling consultant

Jan 2025 – Sep 2025 | [Institute of the Brazilian Savannah](#)

- Hydrological modelling and spatial analysis to prioritize native vegetation protection and restoration in the Cerrado biome.
- Integrated remote sensing, climate time series, land-use maps, Python, and QGIS workflows.

Spatial Data Scientist

Jul 2022 – May 2025 | [UK Centre for Ecology & Hydrology \(UKCEH\)](#)

- Spatial data analysis, management, and mapping for the Water Resources Systems group.
- Catchment attribute development for the [National River Flow Archive \(NRFA\)](#).
- LiDAR processing tools for the [Floods and Droughts Research Infrastructure \(FDRI\)](#).
- Research data curation through the [Environmental Information Data Centre \(EIDC\)](#).

Research Assistant

Jan 2020 – Jun 2022 / *Federal University of Rio Grande do Sul (UFRGS)* in partnership with the *National Water and Sanitation Agency (ANA)*

- Hydrological modelling compatibility with vector river networks.
- National-scale analysis of precipitation, soil moisture, evapotranspiration, and water storage.
- Machine-learning regionalization of reference streamflows and support for the System of Environmental-Economic Accounting for Water (SEEA-Water) in Brazil.

Education

- **PhD in Water Resources**, Federal University of Rio Grande do Sul (UFRGS) (2020 – 2024). Thesis: [Data integration approaches for hydrological analyses in Brazil](#).
- **Master's degree in Water Resources**, Federal University of Rio Grande do Sul (UFRGS) (2018 – 2020)
- **Civil Engineering degree**, Pontifical Catholic University of Rio Grande do Sul (PUCRS) (2011 – 2016)

Tools & Datasets

- **Rating Curve Explorer**: Toolset for organizing, visualizing, and doing initial analysis on streamflow data related to rating-curve derivation.
- **Brazilian Hydrological Availability Data System**: ETL framework for consolidating geospatial hydrological data within ANA's water availability data system, with traceable processing over national hydrographic network versions.
- **Build Hydro Database Brazil**: Tools for retrieving streamflow and rainfall data from stations of the Brazilian Hydrological Network.
- **Reference Mean and Low Streamflow Dataset**: Reference streamflows, including long-term mean flow and q95, for more than 400,000 ungauged river points in Brazil, estimated with machine-learning models.
- **BHO2MGB**: QGIS plugin to process the Brazilian coded river network database in the framework of the MGB large-basin hydrological model.
- **Topographic Position-based Stream definition (TPS)**: Algorithm for drainage extraction on digital elevation models based on topographic position.
- **Attention UNet for shrub segmentation**: Deep-learning pipeline used to identify shrub individuals or clumps in drone-derived RGB imagery.

Publications

- **2026.** [CAMELS-GB v2: hydrometeorological time series and landscape attributes for 671 catchments in Great Britain](#). Gemma Coxon; Yanchen Zheng; **Rafael Barbedo**; Hollie Cooper; Felipe Fileni; Hayley J. Fowler; Matt Fry; Amy Green; Tom Gribbin; Helen Harfoot; Elizabeth Lewis; Germano Gondim Ribeiro Neto; Xiaobin Qiu; Saskia Salwey; Doris E. Wendt. *Earth System Science Data*. derivation of catchment descriptors such as land cover, topography, and rainfall.
Contribution: Contributing author.
- **2025.** [Modeling Long-Term Flow Regimes in Ungauged Basins: An Evaluation of Machine Learning Performance and Uncertainty](#). **Rafael Barbedo**; Iporã Possantti; Mino Viana Sorribas; Rodrigo Cauduro Dias de Paiva; Walter Collischonn. *ESS Open Archive*. evaluation of machine-learning performance and uncertainty for long-term flow regimes in ungauged basins.
Contribution: Main author.
- **2024.** [Data integration approaches for hydrological analyses in Brazil](#). **Rafael Barbedo**. *Doctoral dissertation, Federal University of Rio Grande do Sul*. PhD dissertation.
- **2023.** [A comprehensive strategy for modeling watershed restoration priority areas under epistemic uncertainty](#). Iporã Possantti; **Rafael Barbedo**; Marcelo Kronbauer; Walter Collischonn; Guilherme Marques. *Journal of Hydrology*. drainage-network derivation and water-table analysis.
Contribution: Contributing author.
- **2022.** [How much inundation occurs in the Amazon River basin?](#). Ayan Santos Fleischmann; Fabrice Papa; Alice Fassoni-Andrade; John M. Melack; Sly Wongchuig; Rodrigo Cauduro Dias Paiva; Stephen K. Hamilton; Etienne Fluet-Chouinard; **Rafael Barbedo**; Filipe Aires; Ahmad Al Bitar; Marie-Paule Bonnet; Michael Coe; Jefferson Ferreira-Ferreira; Laura Hess; Katherine Jensen; Kyle McDonald; Alex Ovando; Edward Park; Marie Parrens; Sébastien Pinel; Catherine Prigent; Angélica F. Resende; Menaka Revel; Ake Rosenqvist; Jessica Rosenqvist; Conrado Rudorff; Thiago S.F. Silva; Dai Yamazaki; Walter Collischonn. *Remote Sensing of Environment*. assessment

of inundation extent in the Amazon River basin.

Contribution: Contributing author.

- **2022.** [Water storage variability across Brazil](#). **Rafael Barbedo**; Ayan Santos Fleischmann; Vinícius Siqueira; João Paulo Brêda; Gabriel Matte; Leonardo Laipelt; Alexandre Amorim; Alexandre Abdalla Araújo; Marcus Fuckner; Adalberto Meller; Fernando Mainardi Fan; Walter Collischonn; Anderson Ruhoff; Rodrigo Cauduro Dias de Paiva. *Brazilian Journal of Water Resources*. analysis of water-storage variability across Brazil.
Contribution: Main author.
- **2022.** [Global Evapotranspiration Datasets Assessment Using Water Balance in South America](#). Anderson Ruhoff; Bruno Comini de Andrade; Leonardo Laipelt; Ayan Santos Fleischmann; Vinícius Alencar Siqueira; Adriana Aparecida Moreira; **Rafael Barbedo**; Gabriele Leão Cyganski; Gabriel Matte Rios Fernandez; João Paulo Lyra Fialho Brêda; Rodrigo Cauduro Dias de Paiva; Adalberto Meller; Alexandre de Amorim Teixeira; Alexandre Abdalla Araújo; Marcus André Fuckner; Trent Biggs. *Remote Sensing*. analysis of spatial agreement among evapotranspiration datasets.
Contribution: Contributing author.
- **2022.** [Topographic Position-based Stream definition \(TPS\)](#). **Rafael Barbedo**; Vinícius Siqueira; Walter Collischonn. *Hydrological Sciences Journal*. method to address spatial variability of drainage density in stream networks.
Contribution: Main author.
- **2020.** [Identifying and assessing vegetation behaviour in riparian zones at large scale in the Brazilian Savannah](#). **Rafael Barbedo**. *Master's thesis, Federal University of Rio Grande do Sul*. Master's dissertation.